Diffusion Schrödinger Bridge

Based on "Diffusion Schrödinger Bridge Matching" (2023) by Doucet et al. and related works

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May 1 2026

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Papers for Today

(1) [SDBCD23], “Diffusion Schrödinger Bridge Matching” (2023), Yuyang Shi, Valentin De Bortoli, Andrew Campbell, and Arnaud Doucet. *Note: Scheme was concurrently proposed in [Pel23]*

(2) [DBKMD24], “Schrödinger Bridge Flow for Unpaired Data Translation” (2024), Valentin De Bortoli, Iryna Korshunova, Andriy Mnih, and Arnaud Doucet.

(3) [DBTHD21] “Diffusion Schrödinger bridge with applications to score-based generative modeling” (2021), Valentin De Bortoli, James Thornton, James Heng, Jeremy, and Arnaud Doucet.

Contents

Review of Stochastic Interpolant

Stochastic Interpolant Framework

Many ways to ways to produce the same $(\rho_t, t \in [0,1])$

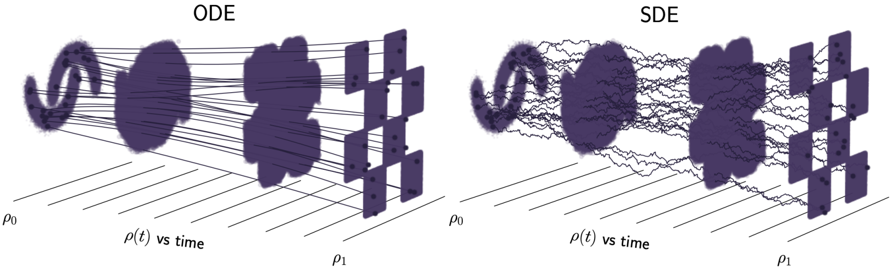


Figure: ODE vs SDE Generative Modeling [ABVE25, Figure 1]

Reciprocal Processes

Recall interpolation: $(x_0, x_1) \sim \nu$ and $z \sim N(0, \mathrm{Id})$ independent of ν

$$x_t = I(t, x_0, x_1) + \gamma(t) z, \quad \rho_t = \mathrm{Law}(x_t)$$

Many processes will have same evolution of marginals!

- Consider specific example

$$x_t = (1-t)x_0 + tx_1 + \sqrt{\varepsilon t(1-t)}\, z$$

- Let $(B_t, t \geq 0)$ be standard Brownian motion independent of (x_0, x_1)

$$(1-t)x_0 + tx_1 + \sqrt{\varepsilon(1-t)}\, B_t$$

- Let $(W_t, t \in [0,1])$ be standard Brownian **bridge** independent of (x_0, x_1)

$$(1-t)x_0 + tx_1 + \sqrt{\varepsilon} W_t$$

Brownian Bridges

- **Conditioning** Brownian motion to start and end at specific point
- $(X_t, t \in [0,1])$ with $X_0 = x$, $X_1 = y$ is Gaussian process with

$$\mathrm{E}\left[X_t\right] = x + t(y - x), \quad \mathrm{Cov}(X_t, X_s) = s(1 - t).$$

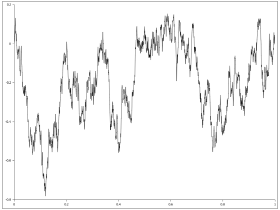


Figure: Sample Brownian Bridge (from Wikipedia)

Reciprocal Process

If $\nu \in \mathcal{P}(\mathbb{R}^d \times \mathbb{R}^d)$ is joint distribution and $(R_{xy}, x, y \in \mathbb{R}^d) \subset \mathcal{P}(C^d[0,1])$ is collection of bridges, write

$$Q = \int_{\mathbb{R}^d \times \mathbb{R}^d} R_{xy} \nu(dxdy).$$

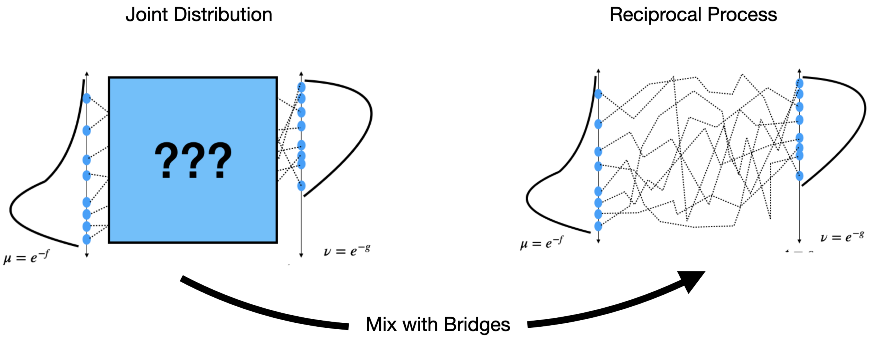


Figure: Reciprocal process: (1) Joint distribution + (2) Bridge Process

Reciprocal Processes vs Markov Processes

Let $(X_t, t \in [0,1])$ denote reciprocal process

- (X_0, X_1) has specified joint distribution
- $(X_t, t \in [0,1]) | X_0, X_1$ has specified bridge distribution

Time Symmetry of Reciprocal Process [LRlZ14]

Let $\varphi : C^d[0,1] \to \mathbb{R}_{\geq 0}$ and $0 \leq s \leq u \leq 1$ then

$$\mathrm{E}[\varphi(X_\cdot) | \mathcal{F}_{[0,s]}, \mathcal{F}_{[u,1]}] = \mathrm{E}[\varphi(X_\cdot) | X_s, X_u]$$

- Compare with **Markov** property: $\mathrm{E}[\varphi(X_\cdot) | \mathcal{F}_{[0,u]}] = \mathrm{E}[\varphi(X_\cdot) | X_u]$

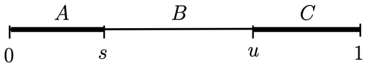


Figure: Reciprocal Process Schematic

Schrödinger Bridge

Choice of Coupling

What's the point of choosing $(X_0, X_1) \sim \nu$? Another **design choice**

- Diffusion models [SSDK+20], one marginal is noise and assume independent
- [AGB+24] analyzes theoretical and practical advantages of ν selection

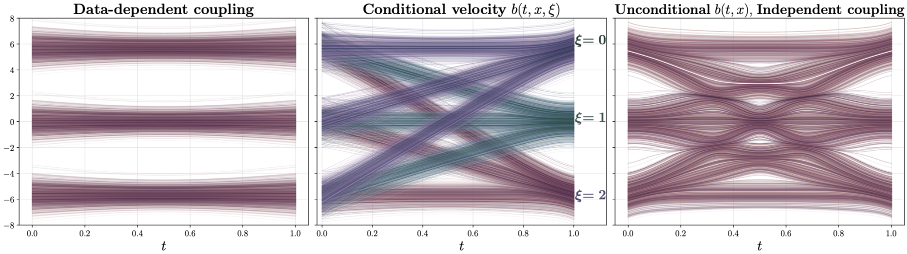


Figure: [AGB+24, Figure 3], Gaussian mixture with 3 modes, different couplings of endpoints

Unpaired Data Translation

How can we find a **natural coupling** between two complicated non-noise distributions?

Figure: Figure 4 in [DBKMD24]

Schrödinger Bridge

Definition

For densities $\mu, \nu \in \mathcal{P}_2(\mathbb{R}^d)$, the **$\varepsilon$-static Schrödinger bridge** from μ to ν is

$$\pi_\varepsilon := \underset{\pi \in \Pi(\mu,\nu)}{\operatorname{argmin}} \left(\int_{\mathbb{R}^d \times \mathbb{R}^d} \frac{1}{2} \|x - y\|^2 d\pi + \varepsilon \mathrm{KL}(\pi | \mu \otimes \nu) \right)$$

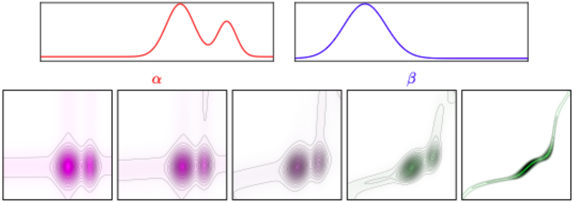


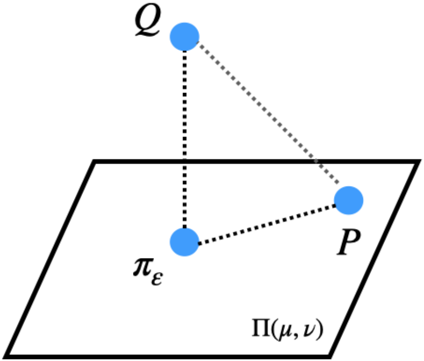
Figure: Figure 4.4 in [PC18]

Schrödinger Bridge

Pythagorean Theorem [Csi75, Theorem 2.2]

Let $Q = \mu(x) r_\varepsilon(x, y)$. For any $P \in \Pi(\mu, \nu)$, it holds that

$$\mathrm{KL}(P|Q) = \mathrm{KL}(P|\pi_\varepsilon) + \mathrm{KL}(\pi_\varepsilon|Q)$$



Dynamic Schrödinger Bridge

- What is the reciprocal process formed from mixing π_ε with ε-Brownian bridges? **Dynamic** Schrödinger bridge
- **Unique** reciprocal process that is also Markov, among fixed choice of intial/terminal marginals and bridge distribution

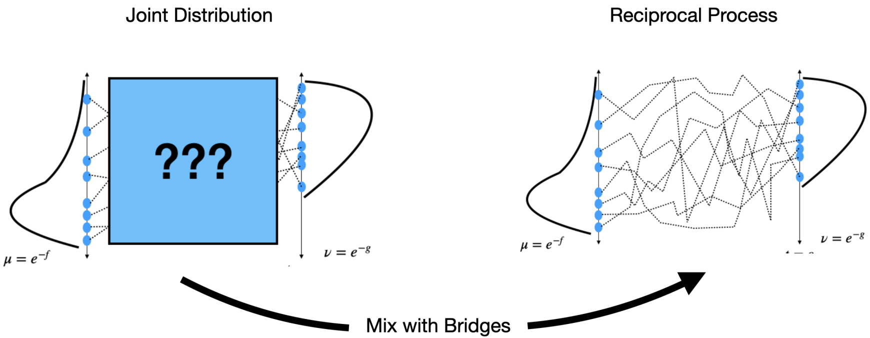


Figure: Dynamic Schrödinger Bridge: (1) Joint distribution π_ε + (2) ε-Brownian Bridges

Iterated Markov Fitting Algorithm

Schematic

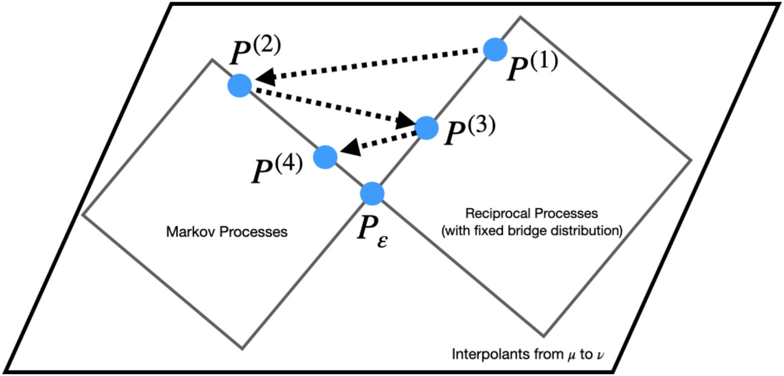


Figure: Schematic for Bridge Fitting-Markov Projection Algorithm

Note: Different from **Sinkhorn algorithm**, another iterative projection algorithm for computing the Schrödinger bridge

Bridge Fitting

- Bridge Fitting = Constructing reciprocal process
- Fix collection of bridge processes for algorithm

Consider samples from Gaussian mixtures, initialize with $P \otimes Q$

$$P = \frac{1}{2}N([-2,-2],\mathrm{Id}) + \frac{1}{2}N([-2,2],\mathrm{Id}), \quad Q = \frac{1}{2}N([2,-2],\mathrm{Id}) + \frac{1}{2}N([2,2],\mathrm{Id})$$

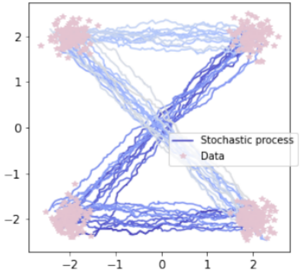


Figure: Bridge Fitting Step, [DBKMD24, Figure 5]

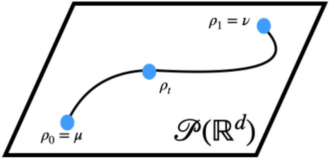
Markov Projection [Gyö86, BS13]

- **Given:** A reciprocal process with marginal flow $(\rho_t, t \in [0,1])$
- **Goal:** Construct **Markov** process with same marginal flow $(\rho_t, t \in [0,1])$

Reciprocal Process

$$dX_t = b_t(X_0, X_t)dt + \sqrt{\varepsilon} dB_t$$

Depends on time at and before t

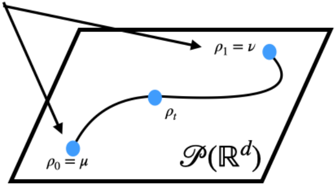


Markov Process

$$dY_t = \mathrm{E}[b_t(X_0, X_t) \,|\, X_t = Y_t]dt + \sqrt{\varepsilon} dB_t$$

Depends only on time t

But all **joint** distributions modified!



Markov Projection

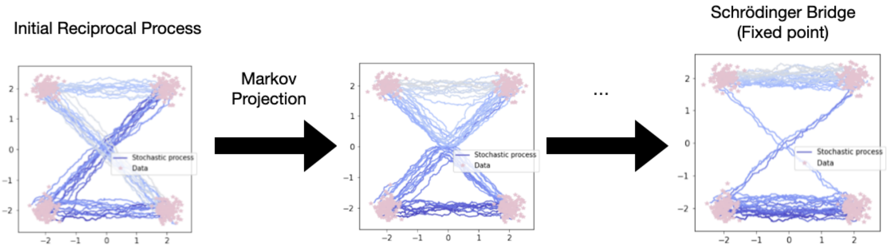


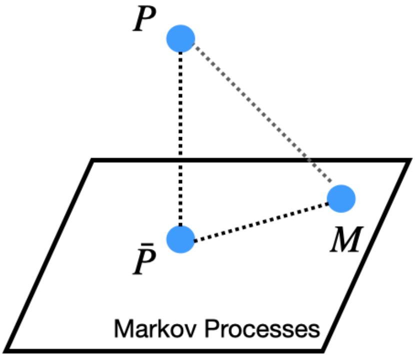
Figure: Iterates of Scheme

Convergence

Pythagorean Theorem of Markov Projections [SDBCD23, Lemma 6]

Fix $P \in \mathcal{P}(C^d[0,1])$ and let $\bar{P} \in \mathcal{P}(C^d[0,1])$ denote its Markov projection. For all $M \in \mathcal{P}(C^d[0,1])$ Markov such that the below quantities are finite,

$$\mathrm{KL}(P|M) = \mathrm{KL}(P|\bar{P}) + \mathrm{KL}(\bar{P}|M).$$



Convergence

Let $\mathcal{R}_\varepsilon$ denote the set of processes with temperature ε Brownian bridges

"Pythagorean Theorem" of Reciprocal Processes

Let $P \in \mathcal{P}(C^d[0,1])$, let $\bar{P}$ denote reciprocal process formed from P with $Q \in \mathcal{R}_\varepsilon$

$$\mathrm{KL}(P|Q) = \mathrm{KL}(P|\bar{P}) + \mathrm{KL}(\bar{P}|Q)$$

Observe that

$$\mathrm{KL}(P|Q) = \mathrm{E}_P\left[\log \frac{dP}{dQ}\right] = \mathrm{E}_P\left[\log \frac{dP}{d\bar{P}}\right] + \mathrm{E}_P\left[\log \frac{d\bar{P}}{dQ}\right]$$

Let $p, q, \bar{p}$ denote the endpoint distributions of $P, Q, \bar{P}$ respectively. By construction, $\bar{p} = p$. Since $\bar{P}, Q \in \mathcal{R}_\varepsilon$,

$$\mathrm{E}_P\left[\log \frac{d\bar{P}}{dQ}\right] = \mathrm{E}_P\left[\log \frac{d\bar{p}}{dq}(x_0, x_1)\right] = \mathrm{E}_p\left[\frac{d\bar{p}}{dq}\right] = \mathrm{E}_{\bar{p}}\left[\frac{d\bar{p}}{dq}\right] = \mathrm{KL}(\bar{P}|Q).$$

Proof of Convergence [SDBCD23, Proposition 7, Theorem 8]

Consider initial reciprocal process $P^{(0)}$, define for $n \geq 0$

$$P^{(2n+1)} := \mathrm{MarkProj}(P^{(2n)}), \quad P^{(2n+2)} := \mathrm{BridgeFit}(P^{(2n+1)})$$

Let P_ε be Schrödinger bridge. By Markov projection Pythagorean Theorem,

$$\mathrm{KL}(P^{(2n)}|P_\varepsilon) = \mathrm{KL}(P^{(2n)}|P^{(2n+1)}) + \mathrm{KL}(P^{(2n+1)}|P_\varepsilon).$$

By the Pythagorean Theorem for reciprocal processes,

$$\mathrm{KL}(P^{(2n+1)}|P_\varepsilon) = \mathrm{KL}(P^{(2n+1)}|P^{(2n+2)}) + \mathrm{KL}(P^{(2n+2)}|P_\varepsilon).$$

Iterating gives for all $N \geq 1$

$$\mathrm{KL}(P^{(0)}|P_\varepsilon) = \mathrm{KL}(P^{(N+1)}|P_\varepsilon) + \sum_{i=0}^{N} \mathrm{KL}(P^{(i)}|P^{(i+1)}).$$

Observe $\mathrm{KL}(P^{(n)}|P^{(n+1)}) \to 0$ as $n \to +\infty$, and $\mathrm{KL}(P^{(n+1)}|P_\varepsilon) \leq \mathrm{KL}(P^{n}|P_\varepsilon)$. Finish with **weak compactness** argument.

Convergence Rate

What about a rate of convergence? Explored in [SCD25]

Convergence Rate Log Concave Marginals [SCD25, Theorem 1]

Let μ, ν be α_μ, α_ν-strongly log concave, respectively. Assume that $(X_0, X_1)|X_t$ is α-log concave for all $t \in [0,1]$ under P_ε.
Let $(P^{(n)}, \geq 0)$ be iterates of scheme, and assume that $\varepsilon > \min(\alpha_\mu^{-1}, \alpha_\nu^{-1})$, then

$$\mathrm{KL}(P^{(n)}|P_\varepsilon) \leq \left(\frac{1}{\varepsilon(\alpha_\mu + \alpha_\nu + \alpha - 2\varepsilon^{-1})} \right)^n \mathrm{KL}(P^{(0)}|P_\varepsilon)$$

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